

Completed results summary

Math Competition Completed Results Submission

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Purpose This report presents 22 proposed M2/M3 contribution families and five focus-hill candidates for the [Open Problems Hack at MIT](#). It brings together exact mathematical scope, formal and computational evidence, comparisons with existing public work and exact source locators for submission review.

How to use this report Read each entry's *Scope*, *Our contribution* and *Prior work and novelty assessment*. References use printed page numbers. Extract the companion ZIP: `README.md` maps the files and `REPRODUCE.md` gives checking commands. Section 6 indexes the claim records and C01–C05/S01–S31 files. If a local link is blocked, follow its printed relative path.

M1 Focus-hill candidates

The **five candidates** distinguish narrow mathematical results from their proposed contribution. **Erdős 3** has a complete standard-axiom certificate for the team's finite separation $79 > 13 \cdot 6$ and sharp cyclic bound of six. **Grothendieck** has standard-axiom proofs for all three-row sign matrices and four named finite matrices. **Matrix multiplication** has standard-kernel certificates that fixing two factor lists determines the third, for all three supplied decompositions, supported by exact source-data checks. Page 2.

Ramsey multiplicity combines an exact weighted construction with a checked count-to-limit bridge. A stronger bound has been publicly announced; its complete proof and publication timing remain under review, so the candidate stays provisional. **BB6** has a Lean kernel proof of an exact 1,235,211-step, 1,519-one halter with no earlier halt, and all six states visited by step 40.

M2 Formalizations of known mathematics

The **10 proposed M2 families** span arithmetic, quaternion scalar definability and discrete geometry. Each subject section distinguishes known mathematics, the proposed formal contribution and existing formal-library overlap. Details, page 6.

M3 Original targets and parent-linked variations

M3A Original open targets and formalized partial advances (10 families)

Complete stated targets. [A100475](#) proves prime-reversal non-ultimate-periodicity from every positive start; [A060957](#) refutes prime-exponent interpolation with arbitrarily long gaps. Open Problem Difficulty Profile (OPDP) problem 43 resolves the historical half-turn miss-two question with a sharp miss-four witness. The Erdős–Lovász result proves the source paper's sharp residual +3 refinement. Exact scopes, credited inputs and bounded prior-work gaps begin on page 12.

Formalized partial advances. Erdős 829's $\log(5/3)$ bound, Erdős 1060's improved constant, the Erdős 944 finite obstruction, A000224's three-prime family, Ferrers deficiency and the Erdős 169 reciprocal bound retain their precise partial scope.

M3B Nontrivial parent-linked variations (2 families)

OPDP13 gives a restricted quartic-exponent classification and quantitative boundary scale. OPDP89 classifies the collapsed-bias likelihood landscape for the four-visible, one-hidden normalized spin restricted Boltzmann machine (RBM), including non-strict local maxima and the exact saddle boundary. Details, page 18.

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1 Contribution overview

This report proposes 27 result families for review: five M1 candidates, 10 M2 formalizations, 10 M3A targets or partial advances, and two M3B variations. These are proposed review units, not accepted credits. Each family groups its main result, supporting theorems and certificates; known inputs and duplicate snapshots are not counted separately.

The handbook categories distinguish the kind of contribution. M1 concerns solutions or formalized advances on the frozen focus set. M2 concerns new, faithful, useful and reusable formalizations of known mathematics. M3A concerns solutions or formalized advances on other eligible original open problems, including newly admitted targets. M3B concerns solutions or formalized advances on nontrivial variations linked to a parent problem. All category assignments remain proposed, with exact scope and prior work assessed entry by entry.

Source labels identify where a problem or statement came from. OPDP means Open Problem Difficulty Profile; OEIS means Online Encyclopedia of Integer Sequences, whose entries use Axxxxxxx identifiers. Numbered Erdős problems and cited research papers provide other sources. Source origin is separate from contribution category.

Reading and checking a result

Start with the exact *Scope*, then *Our contribution* and *Prior work and novelty assessment*. An endpoint is a final named Lean theorem whose statement and axioms have been checked. Each Detailed assessment link opens the corresponding claim record. Section 6 groups the C01–C05 and S01–S31 source, computation, log and receipt locators by M1, M2 and M3. In the extracted companion ZIP, `README.md` provides the file map and `REPRODUCE.md` gives the checking commands and environment requirements. Every printed Detailed assessment path is relative to the extracted ZIP root. If a viewer blocks a local link, open that path from the extracted folder. Page references use the printed page numbers, so the unnumbered cover is not page 1.

2 M1 Focus-hill candidates

These five candidates concern organizer-listed focus hills and present exact restricted results, their mathematical or formal contribution, and their evidence. A narrow obstruction or certified computation can be a partial advance under handbook Section 5; its actual novelty and verification boundary are assessed item by item. Alternative mathematical-progress and formalization routes for the same work form one family.

Detailed assessment links print full paths from the extracted ZIP root. The shorter Detailed card and evidence paths in this section are relative to `Focus_Evidence/`, extracted from `Math_Competition_Completed_Results_Submission_Bundle.zip`. Labels C01–C05 link to the exact source and evidence index on page 21.

2.1 Erdős 3: finite nonoptimality of diagonal pullbacks

Our contribution Our team supplies the 79-point strong-L-free witness at modulus 13, the finite separation $79 > 78$, and the complete formal certificate that every source-defined diagonal pullback of a cyclic four-progression-free subset of $\mathbb{Z}/13\mathbb{Z}$ has size at most 78 and is strictly smaller than the retained witness. The certificate also proves the sharp cyclic four-progression-free maximum of six, linking the witness and obstruction in one contribution.

Scope There is an explicit 79-point set $B \subseteq (\mathbb{Z}/13\mathbb{Z})^2$ containing no configuration $(x, y), (x, y + d), (x, y + 2d), (x + d, y)$ with $d \neq 0$. Every cyclic four-term-progression-free $A \subseteq \mathbb{Z}/13\mathbb{Z}$ has $|A| \leq 6$, and $\{0, 1, 2, 4, 5, 7\}$ attains six. Hence $|B| = 79 > 13|A|$, with $79 > 78$ at the sharp cyclic maximum. Diagonal pullbacks $S_A = \{(x, y) : x - y \in A\}$ are therefore not extremal for this finite strong-L problem. The source-faithful bridge kernel-checks $|S_A| = 13|A|$, the L-free transfer and the strict comparison with B , retaining the original $x - y \in A$ convention.

The valid one-way transfer from a progression-free set to an L-free set remains intact. This result neither determines the strong-L maximum nor yields an asymptotic improvement or a solution or refutation of Erdős 3. The exact equality contradicted here is recorded in the supplied ledger; its status as a recognized obstacle for the parent conjecture is not established.

Verification evidence The cyclic sharpness certificate and source-faithful diagonal-pullback bridge passed Lean 4.33.1 with standard axioms. A fresh seven-module source-only replay on 3 October 2026 checks the actual pullback cardinality, its L-free transfer and the strict comparison with the retained 79-point witness. The original witness and row-partitioned proof are preserved; the Boolean-to-Finset and original-predicate bridges are kernel checked. The source and receipts identify Mathlib `0df444a360eaa60ab8c11dca51a86af692955474`. No native-evaluation axiom or reused project-local compiled object enters this replay.

Prior work and novelty assessment The team's 79-point witness, the separation $79 > 78$ and its completed formal certificate form one finite-obstruction contribution. Peluse's L-configuration work provides relevant external context; the bounded review located no exact primary prior for this finite separation and does not establish world priority. Its relevance as a recognized obstacle for the parent conjecture remains a review question. This is one provisional finite-obstruction M1 candidate; a formalization route would be an alternative assessment of the same family.

Detailed card `cards/FOCUS-E3.md`.

Detailed assessment [Focus_Evidence/cards/FOCUS-E3.md](#).

Formal proof source C01: `CombinedObstruction.lean`, `CyclicSharpness.lean` and `PullbackObstruction.lean`; the complete source chains and correspondence are included.

2.2 Grothendieck: restricted sign-matrix bounds

Our contribution Our team contributes the restricted three-row sign-matrix bound and the four named finite-matrix applications, with their standard-axiom Lean proofs and source-linked checking records. These results provide precise certified cases within the broader sign-matrix problem.

Scope For a sign matrix A , let $C(A)$ be its scalar-sign optimum and $Q(A)$ the supremum of $\sum_{i,j} A_{ij} \langle u_i, v_j \rangle$ over unit-vector families in a specified real inner-product space, with $Q(A) = 0$ for an empty feasible set. For every $3 \times n$ sign matrix and every such space,

$$Q(A) \leq \sqrt{3/2} C(A) \leq \sqrt{2} C(A).$$

Separately, $Q(A) \leq \sqrt{2} C(A)$ is proved for the four explicit matrices `6x6_106`, `6x8_690`, `7x8_53110` and `8x8_1705`. Matrix shape is not the ambient vector dimension. These results concern sign coefficients, not arbitrary real coefficient matrices.

Verification evidence `Row3Granular.lean` preserves the original mathematical body with narrowed imports; both endpoint checks have standard-only axioms. `SemanticTestGranular.lean` verifies the four named matrices with standard-only axioms after import changes and eight arithmetic tactic repairs, preserving the statements and certificate data. Exact source differences, hashes and logs are retained. The certified finite applications are exactly the four matrices named above; no broader orbit classification is claimed.

Prior work and novelty assessment Walsh/Cauchy estimates, semidefinite duality and the CHSH phenomenon are known mathematics. The bounded review did not locate the exact three-row sign-only $\sqrt{3/2}$ theorem or settle priority for the four finite applications; elementary or folklore overlap remains possible. Li’s earlier real-coefficient results through four settings provide relevant prior context. Exact priority remains a separate review question; the source history is preserved alongside the entrant declaration of competition-period production.

Detailed card `cards/FOCUS-GROTH.md`.

Detailed assessment [Focus_Evidence/cards/FOCUS-GROTH.md](#).

Formal proof source C02: `Row3Granular.lean` and `SemanticTestGranular.lean`; endpoint declarations, correspondence and evidence receipts are indexed together.

2.3 Matrix multiplication: one-leg rigidity

A seed is one fixed stored decomposition with factor lists U, V, W . A one-leg change alters only one of those lists. HKS139 is the label used for the supplied decomposition data.

Our contribution Our team contributes the fixed-seed one-leg obstruction: exact rank certificates for all nine paired-factor systems across three seeds, together with standard-kernel proofs of the injectivity and remaining-factor uniqueness statements for all three seeds. The contribution certifies rigidity of these known decompositions under one-leg changes.

Scope Each of three stored HKS139 seeds has rational factor lists U, V, W of shape 23×9 . For each choice of two lists, the 81×23 matrix formed from the rowwise Kronecker products has rank 23. Thus fixing two factor lists uniquely determines the third through the Brent equations: changing only that list cannot reduce its support while preserving the decomposition tensor. Exact rational elimination verifies all nine matrices across the three seeds. The `solution.json` is identical to seed 0 and adds no fourth case.

Verification evidence All three stored HKS139 seeds have standard-kernel certificates for the UV, UW and VW rational injectivity and remaining-factor uniqueness statements. Seed 0 retains its recorded certificate; the two additional seeds passed fresh 71-module Lean 4.33.1 replays on 3 October 2026. Endpoint axioms are only `propext`, `Classical.choice` and `Quot.sound`. The original factors, selected minors, integer inverse tables and all 729 tensor coefficients match their source data. The independent exact rank checks are also retained. The result does not prove a global tensor-rank lower bound, support-139 optimality or impossibility of a simultaneous multi-leg change.

Prior work and novelty assessment The rank-23, support-139 construction is attributed to Smirnov/HKS, and injectivity from a left inverse is standard linear algebra. The team’s contribution is the exact obstruction for these fixed known seeds and its source-linked certificate. The bounded review has not established an earlier identical obstruction, a new mathematical discovery or a nontrivial prior-library delta. Its narrow relevance to the matrix hill is concrete; meaningful novelty remains to be established.

Detailed card `cards/FOCUS-MATRIX.md`.

Detailed assessment [Focus_Evidence/cards/FOCUS-MATRIX.md](#).

Formal proof source C03: `KernelFinal.lean` and the integer row certificates. **Computational certificate** The same index gives all three original seeds, exact rank checks, minors and inverse data.

2.4 Ramsey multiplicity: exact construction and formal lifting

Our contribution Our team contributes the exact weighted 1024-block construction and independently checked monochromatic count, together with the reusable Lean count-to-limit bridge. The bridge treats repeated block labels and diagonal colors, supplies the explicit collision-error bound, and proves the limiting upper-bound implication.

Scope Let c_4 denote the limiting minimum proportion of monochromatic four-cliques in red/blue colorings of a complete graph. The supplied symmetric 1024-block Boolean template has positive integer weights of total $Q = 47,314,248$. Its exact ordered weighted monochromatic K_4 probability is

$$P = \frac{75525646867723584998137743109}{2505750190359427899123170347008} = 0.03014093230773738583 \dots$$

It satisfies the hill reference $P < 20480989/679477248$. For every finite symmetric Boolean template of positive total weight, the generic formal bridge proves $c_4 \leq P$, convergence of finite minimum densities to c_4 , and weighted blow-up error at most $6/(tQ)$ when $t > 0$ and $tQ \geq 4$. Repeated block labels and diagonal colors are included in the count convention.

Verification evidence Independent exact-integer counting reproduces the concrete numerator. `Lifting.lean` and `Limit.lean` passed Lean 4.33.1 checks with standard-only endpoint axioms after four documented compatibility edits that preserve theorem statements. The concrete numerator is verified by exact-integer computation outside Lean. The archived October 2 validation receipt records a nonfinal passed hill check and a Python-plus-lifting trust boundary; its authentication tag was not authenticated in this review.

Prior work and novelty assessment The Technion page for Noam Feinstein’s January 21, 2026 talk announces $c_4 < 0.030139$, a stronger numerical bound in the same normalization. This is a credible stronger prior announcement under review: a complete proof/certificate was not obtained, and the page’s pre-event publication/version has not been authenticated. The team’s construction remains a provisional M1 candidate pending that verification. Lifting, averaging and collision estimates are known mathematics. Mathlib density infrastructure and public lean-flag-algebras uniform blow-up and limit developments provide formal overlap; no matching complete weighted Ramsey interface was located in the bounded comparison. A useful formalization route remains an alternative for this single family. Weaker supplied templates are supporting instances.

Detailed card `cards/FOCUS-RAMSEY.md`.

Detailed assessment `Focus_Evidence/cards/FOCUS-RAMSEY.md`.

Formal proof source C04: `Lifting.lean` and `Limit.lean`. **Computational certificate** The same index locates the exact template, integer-count output and archived validation record.

2.5 BB6: exact six-state halting witness

Our contribution Our team contributes the explicit six-state transition table and a complete Lean kernel certificate for its exact 1,235,211-step blank-tape run, final 1,519 ones, and absence of any earlier halt. A separate endpoint proves that every nonhalting state is visited by step 40; two exact simulators supply independent computational evidence.

Scope On the bi-infinite blank tape, starting in state A at head position zero, the six-state, two-symbol transition table

1RBOLC_1RD1RB_OLH1LE_1LA1LD_OLA1LF_OLEORD

halts after exactly 1,235,211 steps and leaves 1,519 ones. It visits every state A–F by step 40; H denotes halt. The exact simulators additionally record a visited span of 2,028 including the post-halting head position, from -2001 through 26 ; these position/span facts are computational evidence. This is a concrete halting result, not a determination of $S(6)$.

Verification evidence The complete Lean kernel certificate proves exactly 1,235,211 transitions from the blank tape to the halted endpoint, exactly 1,519 one-cells, and no earlier halt. A separate endpoint proves that all six states occur by step 40. The general bit-tape bridge specifies the write-and-move semantics and counts all one-cells, with no truncated-tape assumption. The final `Claims.lean` endpoints depend only on `propext` and `Quot.sound`; no native-evaluation axiom is used. Source hashes, the successful module chain and final axiom output are retained. Two independent exact simulators also reproduce the run. The recovered official validation receipt concerns a different 177,725-step table; no matching organizer acceptance receipt is supplied for this table.

Prior work and novelty assessment The weak consequence $S(6) \geq 1,235,211$ is already superseded by much larger known lower bounds. That comparison does not establish whether this exact machine or its halting result was known. Team ownership is attested by the user. Exact-table priority/equivalence to independently published tables remains unverified. The exact run is directly relevant to the halting-witness hill.

Detailed card `cards/FOCUS-BB6.md`.

Detailed assessment [Focus_Evidence/cards/FOCUS-BB6.md](#).

Formal proof source C05: `Claims.lean`, `Certificate.lean` and `Semantics.lean`, with the complete checked module chain. **Computational certificate** The same index retains the literal table, both simulators and their receipts for the additional head-position and span data.

3 M2 Known mathematics and formalization review

Proof-source labels S01–S31 link directly to included source paths and endpoint lines in the source-locator appendix. Detailed assessment links print full paths from the extracted ZIP root. The shorter Detailed card paths are relative to `Claim_Evidence/`, extracted from `Math_Competition_Completed_Results_Submission_Bundle.zip`. Each card includes the exact verification-record paths and their scope.

The 10 proposed M2 families are organized by mathematical subject. Each entry states its theorem, recorded verification and prior-work limits; its claim card supplies the full declarations, hashes, commands and source comparison. Exact Lean file and endpoint locators are collected on page 22.

Verification environments The four individually documented proposals retain their own checking records below. Six families below use seven independently checked proof files. Their 235 shared source-dependency modules and seven proof units passed the recorded Lean 4.33.1 checks with matching source hashes and standard endpoint axioms. Repeated namespaces and helper names require separate proof-file checks. Version pins and the meaning of each verification record are given in Section 5.

3.1 Number theory and arithmetic

A046969 Conjecture I

Local record complete-a046969-i | [Target source](#)

Our contribution We supply the complete Lean proof of the exact Bernoulli-denominator formula for Conjecture I and its primality corollary, with a checked bridge to the original sequence definition. The five-module Lean 4.33.1 replay verifies the stated prime-pair hypotheses and standard axioms; classical Bernoulli congruences supply the mathematical background.

Scope Write B_{2n} for the $2n$ -th Bernoulli number and den for the positive denominator in lowest terms. For $a(n) = \text{den}(B_{2n}/(2n(2n-1)))$ at $n > 0$, if $p > 3$ and both p and $2p-1$ are prime, then $a(p) = 12(2p-1)$. The original primality conclusion follows. This concerns Conjecture I only; it neither proves Conjecture II nor establishes infinitely many eligible prime pairs.

Verification evidence Independent complete five-module replay on Lean 4.33.1, completed 1 October at 13:09 UTC; exact definition and target correspondence, with standard axioms.

Prior work and novelty assessment Strong classical-overlap risk: the formula follows from classical Bernoulli congruences. Possible M2 formalization review, with final category unresolved. Exact earlier publication and formal-library duplication remain unsettled.

Detailed card `cards/a046969-i.md`.

Detailed assessment [Claim_Evidence/cards/a046969-i.md](#).

Formal proof source S21: A46969.lean (lines 33, 37).

A005258 Apéry prime powers

Local record partial-apery-primewpowers | [Target source](#)

Our contribution We supply a completed prime-power irreducibility formalization, including the general odd non-Wieferich family, its translation and congruence interfaces, and the exact upstream polynomial-definition bridge. The seven-module proofs pass on Lean 4.33.1 and 4.34.1 with standard axioms; the binary and ternary cases remain parts of this single family, alongside the known Eisenstein-based mathematical argument.

Scope For $B_n(X) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k} X^k$ in $Q[X]$, B_{2r} and B_{3r} are irreducible for $r \geq 0$. For every odd prime p , $r \geq 1$ and p^2 not dividing $2^{p-1} - 1$, B_{p^r} is irreducible. The reproduced upstream definition is identified formally.

Verification evidence Seven-module strict replay passes on Lean 4.33.1 and 4.34.1, with matching source hashes and standard axioms.

Prior work and novelty assessment Known-result and formalization-delta review. Earlier public Epoch code already proves the full powers-of-two family; another earlier source sketches the same non-Wieferich odd-prime-power implication. The odd-family formalization may still support one useful M2 proposal after the exact formal delta is assessed; the binary case and overlapping 3-power case do not create extra counts. Sun's book remains uninspected.

Detailed card `cards/apery-odd-formalization.md`.

Detailed assessment [Claim_Evidence/cards/apery-odd-formalization.md](#).

Formal proof source S24: Target5258.lean (lines 31).

A135508 prime index recurrence

Local record partial-a135508 | [Target source](#)

Our contribution We supply the unconditional prime-index gcd criterion, the two sufficient congruence classes, and the more general acquired-divisor theorem, all linked to the original recurrence. The strict Lean 4.33.1 replay and endpoint audits verify these statements with standard axioms; the multiplier-location mechanism is shared with Cloitre’s conditional argument.

Scope Let $x(0) = 0, x(1) = 1$ and $x(n + 1) = 2x(n) + \text{lcm}(x(n), n + 1)$ for $n \geq 1$; for $n > 0$ set $a(n) = x(n + 1)/x(n) - 2$. For prime $p > 3$, $a(p - 1) = p$ exactly when $\text{gcd}(x(p - 3), p - 2) > 1$. For prime $p \geq 7$, either $p \bmod 3 = 2$ or $p \bmod 5 = 2$ suffices. More generally, a divisor $d > 1$ of $x(m)$ and $p - 2$ with $1 \leq m \leq p - 3$ suffices.

Verification evidence Independent strict Lean 4.33.1 replay passed on 2 October at 01:22 UTC, with `autoImplicit` and `relaxedAutoImplicit` both false. All four final axiom reports use only standard axioms. A subsequent six-endpoint audit also explicitly confirms the acquired-divisor theorem uses only standard axioms.

Prior work and novelty assessment Classification-review partial. Cloitre’s full corollary assumes C1, unlike these unconditional subcases, but its proof contains the same multiplier-location mechanism. The contribution is the demonstrated criterion and sufficient families, with that overlap disclosed. No full conjecture or formal infinitude result is claimed.

Detailed card `cards/a135508.md`.

Detailed assessment [Claim_Evidence/cards/a135508.md](#).

Formal proof source S23: A135508.lean (lines 123, 141, 151, 156).

A060841 reciprocal determinant integrality

Our contribution We supply the complete formal bridge from the reciprocal-lcm determinant to its product expression and the finite-and-infinite integrality classification. Recorded Lean 4.33.1 checks verify the exact list through 34 together with 36 and 38 using standard axioms, while the known mathematical classification and existing product/valuation helpers remain credited background.

Scope. For $n \geq 1$, the inverse determinant of the n by n matrix with entries $1/\text{lcm}(i + 1, j + 1)$ is an integer exactly when n belongs to $\{1, \dots, 34, 36, 38\}$.

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment The proof closes the determinant-to-product bridge and the finite and infinite parts of the integrality classification. The mathematical classification was already public, and product/valuation helpers have prior formal overlap. This is a formalization proposal, with no new open-problem claim or claim for the separate denominator conjecture.

Detailed card `cards/m2-research-a060841.md`.

Detailed assessment [Claim_Evidence/cards/m2-research-a060841.md](#).

Formal proof source S29: Lean proof file (lines 1426).

Erdős and Rosenfeld four nearby divisors

Our contribution We supply the explicit eight-consecutive-factor construction proving infinitely many integers have four distinct divisors in the stated interval above their square roots, with constant 40. The construction, distinctness, and all interval inequalities are included in the recorded standard-axiom Lean 4.33.1 verification, yielding a closed proof of this precise lower-bound theorem.

Scope. There are infinitely many natural n with at least four distinct divisors d in the open real interval $(\sqrt{n}, \sqrt{n} + 40n^{1/4})$. The preserved statement uses equivalent floor/ceiling integer endpoints.

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment The eight-consecutive-factor construction includes explicit distinct divisors and every interval inequality. The selected constant is 40. Located stronger benchmark statements and axiomatized material do not supply a closed prior proof, but the search is bounded. No upper-bound or maximality theorem is claimed.

Detailed card [cards/m2-research-erdos887.md](#).

Detailed assessment [Claim_Evidence/cards/m2-research-erdos887.md](#).

Formal proof source S30: Lean proof file (lines 123).

Base three reverse and add carry obstruction

Our contribution We supply the uniform digit-length and carry proof that a palindromic base-three reverse-and-add output forces either palindromic input or carry-free mirrored digit sums. Recorded Lean 4.33.1 verification checks this reusable implication with standard axioms, building on classical carry and symmetry ideas and retaining its single-step scope.

Scope. If x plus its canonical base-3 digit reversal has palindromic base-3 digits, then x already has palindromic base-3 digits or every pair of mirrored input digits sums to less than 3.

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment The formal contribution is a uniform digit-length and carry argument. Classical carry/symmetry context is known; the exact reusable target and its formal-library freshness need review. No perpetual avoidance result for the trajectory from 103 is proved, and MathPages is not claimed to state this exact implication.

Detailed card [cards/m2-research-base3-reverse-add.md](#).

Detailed assessment [Claim_Evidence/cards/m2-research-base3-reverse-add.md](#).

Formal proof source S25: Lean proof file (lines 980).

Theta half sum negative inverse cancellation

Our contribution We supply the direct centered-residue permutation and parity proof of negative-inverse half-sum cancellation. Recorded Lean 4.33.1 verification checks the exact odd-prime, parity, range, and modular-inverse hypotheses with standard axioms, providing a concrete formal treatment of mathematics related to the classical Hardy and Dedekind identities.

Scope. For an odd prime k and odd natural a, b with $1 \leq a, b < k$ and $ab \bmod k = k - 1$, define $H(k, a) = \sum_{j=1}^{(k-1)/2} (-1)^{\lfloor aj/k \rfloor}$. Then $H(k, a) + H(k, b) = 0$.

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment A centered-residue permutation and parity argument prove the cancellation directly. Classical Hardy/Dedekind identities explain the underlying mathematics. The current endpoint retains primality and all parity/range hypotheses; no positivity result for the remaining terms or broader audited hypothesis set is claimed.

Detailed card [cards/m2-research-theta-half-sum.md](#).

Detailed assessment [Claim_Evidence/cards/m2-research-theta-half-sum.md](#).

Formal proof source S26: Lean proof file (lines 61).

Reciprocal square numerator arithmetic

Our contribution We combine equal-minimum 3-adic noncancellation, the reciprocal-square 2-adic exclusion, and a rational-numerator bridge to the stated prime-index divisibility result. The two proof files have recorded standard-axiom Lean 4.33.1 checks; the package supplies these reciprocal-square results while using related ordinary-harmonic valuation infrastructure from Mathlib.

Scope. For every $m \geq 1$, the rational sum $\sum_{k=1}^m 1/k^2$ has nonpositive 2-adic and 3-adic valuations. The selected bridge excludes a prime from its numerator when the valuation is nonpositive. Write $A007406(n)$ for the numerator of $\sum_{k=1}^n 1/k^2$ in lowest terms. Separately, for every prime $p > 3$, $\gcd(p, A007406(p-1)) = p$.

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment The useful package combines equal-minimum 3-adic noncancellation, a 2-adic exclusion and the numerator/prime bridge. Pinned Mathlib already treats ordinary harmonic 2-adic valuations, which is related but different. The full composite- n conjecture and stronger classical Wolstenholme theorem remain outside scope.

Detailed card [cards/m2-research-wolstenholme.md](#).

Detailed assessment [Claim_Evidence/cards/m2-research-wolstenholme.md](#).

Formal proof source S27: Lean proof file (lines 766, 1001); S28: Lean proof file (lines 147, 400).

3.2 Quaternion algebras and scalar definability

OPDP67 quaternion scalar definitions

Local record [m2-opdp67](#) | [Target source](#)

Our contribution We supply the explicit single integer-coefficient equation with two unrestricted quaternion witnesses characterizing scalar elements over every stated ordered commutative ring. The recorded Lean 4.34.1 build and standard-axiom audits include rational and Lipschitz specializations; existing commutation APIs support the construction, while the existential equation provides the concrete formal result.

Scope Over a commutative, linearly ordered strict ordered ring R , a Hamilton quaternion q is scalar exactly when unrestricted quaternions u, v satisfy $((uv - vu)^2 + 4)^2 - (qu - uq)^2 - (qv - vq)^2 = 0$. Specializations define scalar Z in the Lipschitz order and scalar Q in rational quaternions.

Verification evidence Saved Lean 4.34.1 two-witness module build and final integer, rational and general-ring axiom audits passed with standard axioms.

Prior work and novelty assessment Known mathematics and a provisional M2 candidate only. The precise formal-library overlap and reusable formal contribution remain to be established. No formal polynomial compiler, DPRM reduction, full decision-problem classification or solution of $H10(Q)$ is claimed. Variants form one family.

Detailed card [cards/opdp67-quaternion.md](#).

Detailed assessment [Claim_Evidence/cards/opdp67-quaternion.md](#).

Formal proof source S22: Quaternion67TwoWitness.lean (lines 157, 169, 179).

3.3 Discrete geometry

Regular polygon chord and distance multiplicity

Our contribution We supply the coherent regular-polygon formalization linking the chord-length formula and geometric distance classes to the unordered-pair count. Recorded standard-axiom Lean 4.33.1 verification proves the bound by the number of vertices for the stated even polygons, using existing two-circle-intersection uniqueness as supporting infrastructure.

Scope. For every even $n \geq 5$ and every distance d occurring between distinct vertices of the regular n -gon, the number of unordered vertex pairs at distance d is at most n .

Verification evidence Recorded Lean 4.33.1 source verification with standard endpoint axioms; See the Verification environments paragraph at the start of Section 3; these proof files are checked separately because declaration names recur.

Prior work and novelty assessment The proof includes the chord formula and the geometry-to-counting bridge. Generic two-circle-intersection uniqueness already exists in pinned Mathlib, so the final inequality alone does not establish novelty. The contribution concerns the coherent regular-polygon infrastructure and retains even n . General convex extremality, odd n and the parent Erdős94 problem are not claimed.

Detailed card [cards/m2-research-regular-polygon.md](#).

Detailed assessment [Claim_Evidence/cards/m2-research-regular-polygon.md](#).

Formal proof source S31: Lean proof file (lines 982).

4 M3 Original targets and parent linked variations

The twelve proposed M3A/M3B contribution families comprise 10 M3A contributions and two M3B contributions. M3A concerns original open targets and formalized partial advances; M3B concerns nontrivial parent-linked variations. Each record states the precise target, completeness, verification evidence and prior overlap.

4.1 M3A Original open targets and formalized partial advances

A100475 prime reversal cycles

Local record complete-a100475 | [Target source](#)

Our contribution We supply the complete positive-start prime-reversal nonperiodicity proof, joining analytic growth with checked finite exceptions to rule out every periodic tail, including the start one. The independent 84-module Lean 4.33.1 replay verifies the exact original target with standard axioms and source identity.

Scope Let p_n be the n th prime, with $p_1 = 2$, and let $T(n)$ reverse the decimal digits of p_n . Every positive starting value has an orbit under T that is not ultimately periodic. This completely answers the original positive-start question negatively and also covers the start 1.

Verification evidence Independent complete replay on Lean 4.33.1, completed 1 October at 13:03 UTC. All 84 source files match the supplied archive; the pinned target correspondence is recorded. Only standard axioms are reported.

Prior work and novelty assessment Proposed M3A complete-target contribution. No exact prior solution was found in the bounded audit, but ease, folklore and difficulty remain material caveats. Ball–Coxeter pages and the 2004 SeqFan archive remain review gaps. Growth lemmas and sieve modules are not separate claims.

Detailed card [cards/a100475.md](#).

Detailed assessment [Claim_Evidence/cards/a100475.md](#).

Formal proof source S01: A100475.lean (lines 17, 20).

A060957 prime exponent gaps

Local record complete-a060957 | [Target source](#)

Our contribution We supply the complete distinct-factor interpolation refutation and its exact arbitrary-gap strengthening, realizing the eight-level exponent fibre with primes above any prescribed bound. The recorded 45-module Lean 4.33.1 replay and dual-version generalized checks certify the construction with standard axioms and no unproved graph-existence premise, building on known encoding and semigroup-hole methods.

Scope For $P(n)$, the products of subsets of $\{1, \dots, n\}$, the interpolation conjecture is refuted. More strongly, for every gap length $g \geq 1$ and bound B there are a prime $B < p \leq n$, $d > 0$ with p not dividing d , and h such that $p^e d$ lies in $P(n)$ exactly for e in $\{h, \dots, h+3\} \cup \{h+g+4, \dots, h+g+7\}$. This gives g consecutive absent levels between attainable endpoints.

Verification evidence Successful concrete and generalized replay records on Lean 4.34.1 and 4.33.1; the combined pinned 4.33.1 closure passed on 1 October at 16:08 UTC. Exact full-interval target correspondence and standard-axiom reports are retained.

Prior work and novelty assessment Provisional M3A refutation with its strengthening in one logical union. No exact earlier result was located in bounded audits. This does not give the smallest witness or one fixed prime for every g .

Detailed card [cards/a060957.md](#).

Detailed assessment [Claim_Evidence/cards/a060957.md](#).

Formal proof source S02: Main.lean (lines 89, 97); S03: Final.lean (lines 90, 113).

A000224 three prime exclusion

Claim record a000224 | [Target source](#)

Our contribution We supply the all-size exclusion for squarefree products of three distinct odd primes, combining global prime bounds, the exact square-residue-count bridge, and exhaustive quotient/discriminant rejection. The 171-module checks on Lean 4.34.1 and 4.33.1 verify this unbounded family with standard axioms, extending beyond the previously bounded searches and known simpler factorization cases.

Scope For all primes $2 < p < q < r$, put $n = pqr$ and let $a(n)$ count square residues including zero, exactly as in the upstream definition. Then $a(n)(a(n) - 1)$ does not divide $n^2 - 1$. Thus the original equivalence holds on this entire infinite family, with no upper bound on p, q or r .

Verification evidence Complete 171-module checks are recorded on Lean 4.34.1 and 4.33.1. The original 4.34.1 check was incremental; the separate 4.33.1 source replay reused no own compiled artifacts across versions. It changes only three proof tactics in Bounds; 170 module sources are byte-identical. All three final endpoints have standard-only axiom reports in both versions. Core source/log receipts are retrospective, with their original evidence dates retained.

Prior work and novelty assessment Checked infinite partial candidate. The full conjecture and other factorization patterns remain open. Known prime-power and two-prime cases are prior inputs; no exact earlier unbounded pqr result was found in the dated bounded search. See [sources/a000224](#).

Detailed card [cards/a000224.md](#).

Detailed assessment [Claim_Evidence/cards/a000224.md](#).

Formal proof source S04: Target224.lean (lines 33, 43, 50).

Ferrers deficiency hierarchy

Local record partial-ferrers-deficiency44 | [Target source](#)

Our contribution We supply the arbitrary-rank deficiency hierarchy that preserves every supplied optimal factor while adjoining the stated matching colors, with exact later ranks and balanced and one-defect realizations. The 44-source dual-version replays verify the graph/rank transport with standard axioms; known greedy boundary and existential cases remain credited within this broader preservation theorem.

Scope Write α_t for the maximum size of a union of t matchings. An optimal degree- r factor is an edge subset with maximum degree at most r and size α_r . For $1 \leq r \leq k$ and $0 \leq e < r$, a finite Ferrers graph with matching-union ranks $\alpha_r = rk - e$ and $\alpha_{r+1} = (r+1)k - r - e$ has $\alpha_1 = k$. Every supplied optimal degree- r factor is retained while $r - e$ disjoint matching colors of size $k - r$ are appended. For $0 \leq h \leq r - e$, $\alpha_{r+h} = rk - e + h(k - r)$. When $e = 0$ (respectively $e = 1$), pairwise edge-disjoint matching classes can be chosen whose first t classes have union size α_t for every $0 \leq t \leq 2r$ (respectively $0 \leq t \leq 2r - 1$). For $e > 1$, this simultaneous realization of every first- r prefix is not asserted for an arbitrary supplied factor.

Verification evidence 44 modules passed saved strict replays on both Lean 4.33.1 and 4.34.1. Current source hashes match both receipts. Actual Young diagram graph and rank transport is included; standard axioms only.

Prior work and novelty assessment Principal Ferrers partial candidate. It does not resolve the unrestricted Latin Tableau conjecture. The $e = r - 1$ case is old greedy extension; the $r = 3$, $e = 1$ existential scope overlaps McAuley. Arbitrary supplied-factor retention is stronger than that existence claim. Ferrers is not OPDP row 44, which is Kakeya.

Detailed card `cards/ferrers-hierarchy.md`.

Detailed assessment [Claim_Evidence/cards/ferrers-hierarchy.md](#).

Formal proof source S05: `DefectExtension.lean` (lines 104); S06: `YoungDiagramDefect.lean` (lines 8, 42); S07: `YoungDiagramRestrictedPrefix.lean` (lines 158).

Erdős 169 four term reciprocal bound

Local record partial-erdos169-fourap-reciprocal | [Target source](#)

Our contribution We supply the explicit six-residue augmentation of Walker’s base-55 construction and the reciprocal-sum bound exceeding 4.44. The fresh full build and axiom audit verify positivity, absence of nonconstant four-term progressions, and the exact rational-sum inequality for the same finite set; Walker’s digit alphabet and earlier bound remain the credited starting point.

Scope Let $D = \{0, 1, 2, 4, 5, 9, 10, 11, 14, 16, 17, 18, 21, 24, 30, 37, 39, 41, 42, 45, 47\}$, $M = 55^3$, $T_j = \{\sum_{i=0}^{j-1} d_i 55^i : d_i \in D\}$, and $U = \{26286, 26726, 26737, 120061, 120501, 120512\}$. Put $E = T_{22} \cup \bigcup_{u \in U} (u + MT_{21})$ and $A = E + 1$. Then A is positive, has no nonconstant four-term arithmetic progression over $\mathbb{Z}$, and its exact finite reciprocal sum exceeds $111/25 = 4.44$.

Verification evidence Fresh full Lean 4.34.1 release build and verification passed. Positivity, progression exclusion and the rational sum concern the same finite set in one unconditional endpoint. Mathematical source hashes match the release; only the later README differs. Standard axioms only.

Prior work and novelty assessment Principal four-term partial candidate. The base-55 alphabet and earlier bound are Walker’s prior work. Candidate delta is the explicit six-residue augmentation and checked 4.44 bound. This is neither the Erdős 3 divergence problem nor the full Erdős 169 optimum. Earlier certificates are supporting stages.

Detailed card `cards/erdos169-fourap.md`.

Detailed assessment [Claim_Evidence/cards/erdos169-fourap.md](#).

Formal proof source S08: `MainResult.lean` (lines 17).

OPDP43 half turn defect gap

Local record partial-opdp43-halfturn | [Target source](#)

Our contribution We prove that a no-three-in-line set of exactly $2n$ points in an n -by- n grid has half-turn miss count zero or at least four, and certify a ten-point witness attaining four. The clean four-module build verifies the universal theorem and witness geometry with standard axioms, closing the historical miss-two question while crediting the known miss-four examples.

Scope For $A \subseteq \{0, \dots, n-1\}^2$ with $|A| = 2n$ and no three collinear, put $R(x, y) = (n-1-x, n-1-y)$. The one-sided miss count $|\{p \in A : R(p) \notin A\}|$ is either 0 or at least 4. A checked ten-point witness on the 5 by 5 grid has miss exactly 4. Symmetric-difference cardinality is twice the one-sided miss.

Verification evidence Saved clean Lean 4.34.1 release build passed with all source hashes matching and standard axioms. The universal endpoint includes $n = 0, 1, 2$, and the witness geometry is kernel-checked.

Prior work and novelty assessment The Flammenkamp symmetry page explicitly asks whether half-turn miss 2 can occur and records known miss 4. This theorem answers that subsidiary question completely; it is only partial progress on the broader no-three-in-line problem. Elementary or folklore overlap remains for review. No claim covers sharpness at every n or a new $n = 75$ construction.

Detailed card [cards/opdp43-halfturn.md](#).

Detailed assessment [Claim_Evidence/cards/opdp43-halfturn.md](#).

Formal proof source S14: Main.lean (lines 8, 56).

Erdős 1060 uniform quantitative partial bound

A formal partial bound toward Erdős 1060. The full zero-constant and polylogarithmic goals remain open.

Our contribution We use the retained powerful-core size constraint and a weighted generating function to reduce the uniform counting coefficient below $\log(3)/3$ to the stated constant C . The fresh ten-module Lean 4.33.1 replay verifies the canonical count, all asymptotic hypotheses, and the strict improvement with standard axioms, retaining the prior squarefree injection and unrestricted-core bound as inputs.

Exact result and evidence Scope. Let $\sigma(k)$ be the sum of positive divisors of k , and let $f(n)$ count positive integers k satisfying $k\sigma(k) = n$. With $C = \log(2539/1000)/3 + \log(10/9)/2$, for every $\varepsilon > 0$ there is N such that every $n \geq N$ satisfies $f(n) \leq \exp((C + \varepsilon) \log(n) / \log(\log(n)))$. The formal count is exactly the finite set $\{k \leq n : k\sigma(k) = n\}$. The strict comparison $C < \log(3)/3$ is also proved.

Prior work and novelty assessment The retained powerful core satisfies $P^2 \leq n$. A weighted generating-function bound with $x = 9/10$ improves the located general unrestricted-core constant from $\log(3)/3$ to $C = 0.363270358736427\dots$. All arithmetic and asymptotic hypotheses are discharged in the final canonical-count declarations. Scott Duke Kominers, On the Number of Solutions of $k\sigma(k) = n$, 2026 working paper, presents the unrestricted-core bound with $\log(3)/3$ and attributes squarefree injectivity to Noppakaew–Pongsriiam. Those arguments are prior mathematics.

Verification evidence. Fresh isolated Lean 4.33.1 compilation passed all 10 own source modules, with no own olean reuse. Before/after source hashes agree and final canonical theorem types and axiom lists were printed. Only propext, Classical.choice and Quot.sound occur.

Scope limits. No solution of either full Erdős 1060 goal. No new claim for the known square-free injection or unrestricted-core encoding. No formalized explicit $O(\text{error-rate})$ refinement or optimized-majorant sharpness. No lower bound for actual fiber counts, odd-target improvement or certified novelty. Current official forum discussion reads returned HTTP 403; bounded searches do not certify priority.

Attribution. Copied canonical Formal Conjectures reference preserves its Apache-2.0 notice; full license is in notices/erdos1060-formal-conjectures-Apache-2.0.txt. Kominers prior-work attribution is retained. No blanket new license, author endorsement or release grant is inferred.

Detailed card [cards/erdos1060-uniform-partial.md](#).

Detailed assessment [Claim_Evidence/cards/erdos1060-uniform-partial.md](#).

Formal proof source S15: SigmaFiberAsymptotic.lean (lines 38, 49); S16: EntropyConversion.lean (lines 45).

Erdős Lovász sharp residual cover bound

Complete proof of the explicit +3 refinement asked after Lemma 2 of arXiv 2606.24878v2.

Our contribution This result closes the source paper’s explicit +3 residual-cover question through refined colored-link budgets and equality-case exclusions, then derives the all-r bound with additive constant three. The isolated 18-module Lean 4.34.1 replay and separate triangle check certify the result and sharpness, building on the credited matching infrastructure and +4 theorem.

Exact result and evidence Scope. Write $\tau(J)$ for the minimum number of vertices meeting every edge, and $|J|$ for the number of edges. For every finite intersecting r -uniform hypergraph J of maximum vertex degree at most 3, $4\tau(J) \leq |J| + r + 3$. Degree-four peeling yields the same inequality without a degree restriction and hence $3r \leq |H| + 3$ whenever $\tau(H) = r$; for $r = 8$ this gives $|H| \geq 21$. The additive constant 3 is optimal, with a separately checked known triangle example.

Prior work and novelty assessment The source paper’s residual bound has +4 and explicitly asks for +3. Refined colored-link edge budgets and finite equality-case exclusions establish +3, and the same family improves its explicit all-r bound from $3r-4$ to $3r-3$. Varun Sivashankar, An Improved Lower Bound for the Erdős–Lovász Cover Number Problem, arXiv 2606.24878v2, 29 July 2026 supplies the unconditional matching/witness infrastructure, +4 theorem and known sharpness examples. The paper’s stronger asymptotic coefficient about 3.053 is prior work and is not improved. Its ancillary source prefix is reused with explicit attribution and only the documented Lean 4.34.1 compatibility changes. The entire Kahn-dependent asymptotic section is excluded.

Verification evidence. An independent isolated Lean 4.34.1 replay passed all 18 Main-closure modules. The proof-output directory began empty, source hashes stayed unchanged, nine external revisions were pinned and clean, and the separate endpoint-definition audit passed. Final endpoints use only `propxt`, `Classical.choice` and `Quot.sound`; no Kahn hypothesis, hidden packing lemma or extra mathematical assumption is present. The known triangle witness was separately checked against the rebuilt core, outside the 18-module replay.

Scope limits. No resolution of the original $O(r)$ Erdős 21 question, already solved by Kahn. No determination of $g(r)$ or improvement of the source paper asymptotic coefficient $(41 - \sqrt{19})/12$. No separate claim for `lower_three_sub`, $r = 8$, triangle sharpness or infrastructure. No certified first-publication priority. Current official forum discussion reads returned HTTP 403; bounded searches do not certify priority.

Attribution. Original 114,124-byte ancillary `Theorem1.lean` is preserved with SHA256 `f494b2570a7be0b42445a54aed61662bffffd938df13ff9a19a97e54781dfcb20`. The arXiv source submission displays CC BY 4.0 <https://creativecommons.org/licenses/by/4.0/>; the ancillary file has no separate software-license header. That provenance is preserved without inventing a different license. AuthorBaseline records reuse and compatibility changes; no author endorsement or blanket release grant is inferred.

Detailed card `cards/erdos21-sharp-residual.md`.

Detailed assessment [Claim_Evidence/cards/erdos21-sharp-residual.md](#).

Formal proof source S17: `Main.lean` (lines 25, 33, 41, 49, 56).

Erdős 944 finite order obstruction

Our contribution We supply the stronger finite order obstruction for vertex-critical four-chromatic graphs with the stated edge resilience, including the quadratic and integer independent-set filters. The isolated Lean 4.34.1 replay and target audit verify all three inequalities with standard axioms, improving the explicit finite comparison while preserving the prior stronger eventual asymptotic result.

Necessary order bounds For a finite simple graph G with n vertices and an integer $r \geq 1$, assume its chromatic number is four, deleting any vertex lowers it, and deleting any set of at most r edges does not lower it. Then $27r + 19 \leq 4n$. The same hypotheses give $27(r+1)(n-1) \leq (2n+1)^2$ and $9b(r+1) \leq (2b+1)(n-b)$, where $b = \lfloor (n+1)/3 \rfloor$. All three bounds form one family.

Prior work and novelty assessment The clean inequality strengthens the explicit $n \geq 5r + 6$ finite obstruction in Skottova and Steiner, Proposition 5.1. Stronger quadratic and integral independent-set filters are also proved; for $r = 4$, the integer filter requires $n \geq 33$ rather than the linear bound alone requiring $n \geq 32$. Skottova and Steiner’s separate asymptotic estimate $r = O(n/(\log n)^c)$, for some absolute $c > 0$, is stronger eventually and is not improved here. The public $r = 1$ construction is prior work.

Verification evidence Independent isolated Lean 4.34.1 replay compiled three proof modules and the separate `TargetAudit.lean` from unchanged frozen sources. Its own output directory began empty and earlier project proof caches were excluded; pinned third-party caches were reused. Exact target definitions and conjunction were audited. Every final endpoint uses only `propext`, `Classical.choice` and `Quot.sound`. The immutable source archive and all source/log hashes match the independently checked package; assembly is an identity check, not another Lean run.

Limits and attribution This is a finite $k = 4$ necessary obstruction. It supplies no existence construction, no full Erdős 944 solution, and no statement for arbitrary infinite vertex types or $k > 4$. Classical coding methods and the reused Formal Conjectures definitions are attributed, with a separate Apache-2.0 notice. Novelty and official eligibility remain separate review questions.

Detailed card `cards/erdos944-order-bound.md`.

Detailed assessment [Claim_Evidence/cards/erdos944-order-bound.md](#).

Formal proof source S18: `Erdos944Bounds.lean` (lines 28, 42, 54).

Erdős 829 completed formal bound

Our contribution We combine the finite weighted-code certificate, all-target gcd transfer, and prime cutoff to prove the uniform $\log(5/3)$ bound for ordered nonnegative cube-sum representations. The fresh 33-proof-module replay and two audits verify the exact unrestricted target count with standard axioms, building on prior ratio-root, congruence, and coding methods.

Exact formal endpoint Let $R(n)$ count ordered nonnegative integer pairs (x,y) with $x^3+y^3 = n$, including zero coordinates. For every real $\varepsilon > 0$ there is a natural N such that every natural $n \geq N$ satisfies $R(n) \leq \exp((\log(5/3) + \varepsilon) \log n / \log \log n)$. The final count correspondence is checked, and there is no primitive-pair, cubefree, root-count or encoding hypothesis left in the endpoint. The threshold N is existential, not a displayed numerical cutoff.

Verification evidence The rebuilt closure passed an isolated build and a separately orchestrated fresh replay on Lean 4.33.1 with Mathlib `0df444a360eaa60ab8c11dca51a86af692955474`. All 33 theorem modules and two semantic audit modules exited successfully; owned proof outputs began empty and

only pinned third-party caches were reused. Final endpoint axioms are `propext`, `Classical.choice` and `Quot.sound`. Of the 33 theorem modules, 28 match historical source hashes and five are reconstructed or repaired variants now covered by these new checks: `CubicRoots`, `CubicRatios`, `CubicDeterminant`, `CubicSeparation` and `CubicEncoding`. The checked source, commands, logs and hashes are preserved.

Prior work and novelty assessment The verified coefficient $\log(5/3)$ improves the elementary $\log 2$ comparison. Merikoski’s ratio-root injection and determinant congruence, and classical Elias/Plotkin localization, are prior inputs. A bounded primary-theorem comparison did not find an inspected height/rank or Thue consequence subsuming the stated all-target coefficient; it does not certify novelty or a best-known bound.

Scope and completeness The theorem counts ordered nonnegative pairs rather than arbitrary signed pairs. It remains much weaker than the conjectured polylogarithmic bound and does not solve Erdős 829. The checked $\log(5/3)$ endpoint is one precisely scoped partial-result family.

Detailed card [cards/erdos829-log-five-thirds.md](#).

Detailed assessment [Claim_Evidence/cards/erdos829-log-five-thirds.md](#).

Formal proof source S20: `FiveThirdsCanonical.lean` (lines 8, 14).

4.2 M3B Nontrivial parent linked variations

OPDP13 and OPDP89 are proposed as nontrivial parent-linked variations, with the exact quantitative or restricted classification targets stated below.

OPDP13 quartic exponent classification

Local record `partial-opdp13-quartic` | [Target source](#)

Our contribution We supply the exact exceptional-exponent classification, cubic transition, and sharp logarithmic boundary scale with leading constant one for the least positive admissible exponent. The recorded 40-proof-module Lean 4.34.1 build verifies the function-level squared-norm statements with standard axioms, refining the known quartic family, radical threshold, and qualitative eventual positivity.

Scope For $H_t(z) = (1 + |z|^2)^4 - t|z|^4$ and $6 < t \leq 7$, a positive integer power H_t^N is a finite sum of squared absolute values of holomorphic polynomials exactly when $N = 2, N \geq 4$, or $N = 3$ with $924 - 210t + 18t^2 - t^3 \geq 0$. For $6 < t < 8$ every admissible N satisfies $12N(8 - t) \geq t(10 - t)$. For $7 \leq t < 8$ it also satisfies $81Nt^{4N} \leq 614656(16 - t)^{4N}$.

Verification evidence Saved standalone final Lean 4.34.1 build passed: 40 proof modules plus the aggregate, with current source hashes matching. Final endpoint axiom logs report only standard axioms.

Classification and limits One OPDP13 partial family. Write $\delta = 8 - t$. For every $\varepsilon > 0$, sufficiently small δ forces admissible N to obey $\delta N \geq (1 - \varepsilon) \log(1/\delta)$, while every integer $N \geq (1 + \varepsilon) \log(1/\delta)/\delta$ has strictly positive coefficients and a squared-norm representation. The defined least admissible positive index has normalized ratio $\delta \cdot \text{index} / \log(1/\delta)$ between $1 - \varepsilon$ and $1 + \varepsilon$. This is a checked epsilon statement; no separate filter-limit declaration is claimed.

Prior work and novelty assessment The family, the $t < 8$ threshold, the $t = 7$ square and qualitative eventual positivity are known. Candidate additions are the finite-exception classification and matching quantitative boundary scale with leading constant 1. The accessible Tan-To appendix concerns a different stabilizing family and rate; the main effective theorem remains an uninspected overlap lead. The claim is the stated quartic family and does not solve general OPDP13.

Detailed card [cards/opdp13-quartic.md](#).

Detailed assessment [Claim_Evidence/cards/opdp13-quartic.md](#).

Formal proof source S09: Evaluation.lean (lines 116); S10: Bound.lean (lines 86); S11: Growth.lean (lines 59); S12: Function.lean (lines 9); S13: LeastIndex.lean (lines 40).

OPDP89 sharp four parity classification

Our contribution We supply the sharp collapsed-fibre local-maximum threshold for the four-visible, one-hidden pure-parity RBM, including stationarity and both signs of variation at and beyond the boundary in every full parameter neighborhood. The fresh 17-module replay and semantic audit verify this higher-order classification with standard axioms, extending the known singular-stationarity setting.

Exact likelihood landscape For $s \in \{-1, 1\}^4$, $b, w \in \mathbb{R}^4$, and $c \in \mathbb{R}$, let the pure four-parity data be $Q_\rho(s) = (1 + \rho \cdot s_1 \cdot s_2 \cdot s_3 \cdot s_4)/16$ with $0 < |\rho| \leq 1$. Consider the normalized four-visible, one-hidden spin RBM $P_{b,c,w}(s)$ proportional to $\exp(b \cdot s) \cdot \cosh(c + w \cdot s)$, and its expected log likelihood ℓ_ρ . At every collapsed point $(0, c_0, 0)$, with c_0 any finite real, every affine parameter-direction derivative is zero. It is a local maximum in the full nine-dimensional parameter space if and only if $\tanh(c_0)^2 < (3 + 2|\rho|)/(3 + 6|\rho|)$. At equality or beyond, every full parameter neighborhood has both a strictly higher and a strictly lower likelihood point. Every collapsed point has a strictly better finite RBM somewhere in parameter space. Distinct equal-likelihood points occur in every neighborhood, so the local maxima are non-strict.

What the extension adds The classification includes both the local-maximum region and the saddle boundary along the finite collapsed hidden-bias fiber. Stationarity holds in every affine parameter direction. The saddle conclusion gives both strict signs in every full neighborhood, including the equality boundary.

Verification evidence All 17 frozen source modules passed fresh isolated Lean 4.33.1 revalidation with Mathlib 0df444a360eaa60ab8c11dca51a86af692955474, in 108.552 seconds. Own proof outputs began empty; no old own olean files were reused, and pinned third-party caches were reused. Before and after hashes match; all module exit codes are zero, with no warnings or errors. The final 41 printed axiom declarations contain only propext, Classical.choice and Quot.sound. A separate semantic type audit also passed.

Prior work and novelty assessment The theorem is restricted to one hidden unit and pure four-parity data. No distribution-local maximum, strict optimization trap, attraction basin, EM behavior or training-time bound is proved. Mixture critical-line and splitting-Hessian results are prior work; the broad OPDP89 program is not fully resolved.

Detailed card [cards/opdp89-parity-local-maxima.md](#).

Detailed assessment [Claim_Evidence/cards/opdp89-parity-local-maxima.md](#).

Formal proof source S19: SharpClassification.lean (lines 22, 33, 40).

5 Verification evidence and source provenance

5.1 Verification standards

Every retained entry states a completed result at its exact scope, its contribution, prior work and the verification actually recorded. A completed partial theorem is a result in its own right; it is not a claim that its parent open problem has been solved. Category placement, novelty and competition acceptance remain separate assessments.

Formal results retain their exact Lean sources, endpoint declarations, source hashes and successful build or axiom records. Standard foundations mean `propext`, `Classical.choice` and `Quot.sound` as applicable to the endpoint; BB6 uses only `propext` and `Quot.sound`. Each receipt applies to its stated source closure and environment. Packaging performs source-identity checks, not a new mathematical proof run.

Exact computational certificates are labeled separately. Ramsey’s concrete numerator retains exact computational evidence outside Lean. The matrix one-leg results for all three seeds now have standard-kernel certificates as well as exact computational checks. Each recorded verification method applies only to its stated endpoint.

5.2 Source identity and attribution

All 27 entries have a detailed card and local source/evidence pointers in this bundle. The entrants declare that all supplied results are their team’s work produced during the competition. Original source and replay timestamps are preserved as artifact history. The team owns the supplied mathematical, computational and formal work; independently credited mathematical inputs, library code and source-license notices remain attributed. Source hashes and archive records establish artifact identity, not world priority or an authenticated discovery date.

The proof records retain their actual environment pins: Lean 4.33.1 with Mathlib `0df444a360eaa60ab8c11dca51a86af692955474`, or Lean 4.34.1 with Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`, as stated per result. The bundle includes the complete project-local proof sources for the listed endpoints, source hashes and replay instructions. The pinned Lean compiler and third-party Mathlib packages are installed separately.

6 Proof and verification files

The appendix groups all 27 result families by proposed category. Stable labels C01–C05 and S01–S31 identify the same proof and evidence files used in the main text. The 22 M2/M3 families have 56 endpoint declarations in 31 distinct Lean files; their complete statements, line numbers, SHA-256 hashes and verification evidence paths are recorded in `Claim_Evidence/claim-navigation-22.json`. The five M1 candidates have corresponding source and evidence records in `Focus_Evidence/focus-candidates.json`. Paths for M2 and M3 below are relative to the extracted submission bundle. The accompanying `REPRODUCE.md` gives the pinned environment and replay commands. `Reproduction_Sources/SOURCE_CLOSURE.json` maps each original-family endpoint to its included project-local module dependencies, build order and preserved source hashes. Detailed cards and the two evidence indexes identify the corresponding exact computations, build logs and receipts.

6.1 M1 Focus-hill candidates

The C01–C05 paths below are relative to `Focus_Evidence/` in `Math_Competition_Completed_Results_Submission_Bundle.zip`. The accompanying `focus-candidates.json` records all endpoint names, line numbers, source hashes and evidence paths; `source-origins.json` preserves the original source correspondence. Extract the complete submission ZIP to preserve the relative card, source and evidence links.

C01: Erdős 3: finite diagonal-pullback nonoptimality

Detailed card and assessment `cards/FOCUS-E3.md`.

Formal proof source `evidence/FOCUS-E3/complete-certificate/CombinedObstruction.lean`; `explicit_L_witness_beats_every_cyclic_AP_free_set` (line 19).

Formal proof source `evidence/FOCUS-E3/complete-certificate/CyclicSharpness.lean`; `cyclic_four_AP_extremum_is_six` (line 12).

Formal proof source `evidence/FOCUS-E3/complete-certificate/CyclicUpper.lean`; `cyclic4Free_card_le_six` (line 11).

Formal proof source `evidence/FOCUS-E3/complete-certificate/R17SplitReplay.lean`; `B_card` (line 18); `B_Lfree` (line 72).

Formal pullback bridge `evidence/FOCUS-E3/pullback-bridge/source/Fugu3PullbackCore.lean`; `card_SA` (line 63). `evidence/FOCUS-E3/pullback-bridge/source/PullbackObstruction.lean`; `card_diagonal_pullback_le_78` (line 31); `retained_witness_beats_every_diagonal_pullback` (line 38).

Bridge replay and source correspondence `evidence/FOCUS-E3/pullback-bridge/fresh-replay/build-receipt.json`; `evidence/FOCUS-E3/pullback-bridge/SOURCE_CORRESPONDENCE.json`; portable commands in the adjacent `README.md`.

Verification evidence `evidence/FOCUS-E3/R17SplitReplay-receipt.json`; `evidence/FOCUS-E3/complete-certificate/CyclicSharpness-receipt.json`; `evidence/FOCUS-E3/complete-certificate/build-receipt.json`. Full logs, environment pins and source correspondence are indexed in the detailed card.

C02: Grothendieck: restricted sign-matrix bounds

Detailed card and assessment `cards/FOCUS-GROTH.md`.

Formal proof source `evidence/FOCUS-GROTH/Row3Granular.lean`; `row3_Q_le_sqrt_three_halves_mul_C` (line 397).

Formal proof source `evidence/FOCUS-GROTH/SemanticTestGranular.lean`; `semantic_6x6_106` (line 276); `semantic_6x8_690` (line 311); `semantic_7x8_53110` (line 346); `semantic_s8x8_1705` (line 381).

Verification evidence `evidence/FOCUS-GROTH/`. Full logs, environment pins and source correspondence are indexed in the detailed card.

C03: Matrix multiplication: one-leg rigidity

Detailed card and assessment cards/FOCUS-MATRIX.md.

Formal proof source evidence/FOCUS-MATRIX/kernel/KernelFinal.lean; rational_injective_uv (line 103); rational_injective_uw (line 142); rational_injective_vw (line 181); remaining_factor_unique_uv (line 192); remaining_factor_unique_uw (line 204); remaining_factor_unique_vw (line 216).

Additional formal proof source evidence/FOCUS-MATRIX/additional-seeds/seed_1/proof/Seed1KernelFinal.lean; the three injectivity endpoints (lines 103, 142, 181) and three uniqueness endpoints (lines 192, 204, 216). Fresh build, source-data correspondence and axiom receipts are included alongside the source.

Additional formal proof source evidence/FOCUS-MATRIX/additional-seeds/seed_2/proof/Seed2KernelFinal.lean; the three injectivity endpoints (lines 103, 142, 181) and three uniqueness endpoints (lines 192, 204, 216). Fresh build, source-data correspondence and axiom receipts are included alongside the source.

Verification evidence evidence/FOCUS-MATRIX/kernel/kernel-check-receipt.json; evidence/FOCUS-MATRIX/matrix-compiled-receipt.json; evidence/FOCUS-MATRIX/matrix-rigidity-exact-recheck.json. Full logs, environment pins and source correspondence are indexed in the detailed card.

C04: Ramsey: exact weighted construction and formal lifting

Detailed card and assessment cards/FOCUS-RAMSEY.md.

Formal proof source evidence/FOCUS-RAMSEY/formal/Lea/RamseyLifting/Lifting.lean; density_sub_P_abs_le (line 1018); lifting_bound (line 1117); density_tendsto (line 1127).

Formal proof source evidence/FOCUS-RAMSEY/formal/Limit.lean; c4_le_Pval (line 536).

Verification evidence evidence/FOCUS-RAMSEY/exact-count/independent_validation.json; evidence/FOCUS-RAMSEY/formal/replay.json. Full logs, environment pins and source correspondence are indexed in the detailed card.

C05: Busy Beaver 6: exact halting witness

Detailed card and assessment cards/FOCUS-BB6.md.

Formal proof source evidence/FOCUS-BB6/kernel/Claims.lean; exact_halting_witness (line 14); visits_all_six_states (line 22).

Formal proof source evidence/FOCUS-BB6/kernel/Semantics.lean; moveR_written (line 11); moveR_left_tail (line 15); moveR_right (line 20); moveL_left (line 22); moveL_current (line 24); moveL_written (line 28); moveL_right_tail (line 32); no_earlier_halt (line 43); ones_eq_count_of_strip_zero (line 74).

Verification evidence evidence/FOCUS-BB6/bb6-strongest-replay.json; evidence/FOCUS-BB6/kernel/build-receipt.json; evidence/FOCUS-BB6/kernel/final-certificate-receipt.json; evidence/FOCUS-BB6/kernel/final-check-receipt.json. Full logs, environment pins and source correspondence are indexed in the detailed card.

6.2 M2 Known mathematics and formalizations

Source and claim	Included source and endpoint lines
S21 a046969-i	Claim_Evidence/proofs/a046969-i/A46969.lean Endpoint lines: 33, 37
S22 opdp67-quaternion	Claim_Evidence/proofs/opdp67/OpHack/Quaternion67TwoWitness.lean Endpoint lines: 157, 169, 179
S23 a135508	Claim_Evidence/proofs/a135508/A135508.lean Endpoint lines: 123, 141, 151, 156
S24 apery-odd-formalization	Claim_Evidence/proofs/apery-odd/OpHack/AperyPrimePower/Target5258.lean Endpoint lines: 31

Source and claim	Included source and endpoint lines
S25 m2-research-base3-reverse-add	Claim_Evidence/proofs/m2-research/claims/proofs/d8fetc50c50b01cf689691ca19ede76b3714975c8cc895d8a151a13ec01070c9.lean Endpoint lines: 980
S26 m2-research-theta-half-sum	Claim_Evidence/proofs/m2-research/claims/proofs/1948663a1348321db557d85cd4e05e9b3b1ad9e03a8dc3f7a342c17afc215ca3.lean Endpoint lines: 61
S27 m2-research-wolstenholme	Claim_Evidence/proofs/m2-research/claims/proofs/30761d1c63ddf3a50fc6ee930e68dadb2cd63742913abfd9007567ec00f72c8e.lean Endpoint lines: 766, 1001
S28 m2-research-wolstenholme	Claim_Evidence/proofs/m2-research/claims/proofs/87b6dd5bd86142923b09c0207d04800a82e423eff3fe4178f53bb2a873a8db5e.lean Endpoint lines: 147, 400
S29 m2-research-a060841	Claim_Evidence/proofs/m2-research/claims/proofs/5e4bb2ede244d1de75b753bc0115a7173872d8b4ecb488d6e6e94828e2691d28.lean Endpoint lines: 1426
S30 m2-research-erdos887	Claim_Evidence/proofs/m2-research/claims/proofs/269a2caa15a591595c54001b8cc5360c677ff25f63904e9017548e7492cf965f.lean Endpoint lines: 123
S31 m2-research-regular-polygon	Claim_Evidence/proofs/m2-research/claims/proofs/cc13704faafdd146113fbb944434a585102dde0dba7700cccacc9f194ba56777.lean Endpoint lines: 982

6.3 M3 Original targets and parent linked variations

Source and claim	Included source and endpoint lines
S01 a100475	Claim_Evidence/proofs/a100475/A100475.lean Endpoint lines: 17, 20
S02 a060957	Claim_Evidence/proofs/a060957/OpHack/SubsetProducts/Main.lean Endpoint lines: 89, 97
S03 a060957	Claim_Evidence/proofs/a060957/OpHack/SubsetProducts/Generic/Final.lean Endpoint lines: 90, 113
S04 a000224	Claim_Evidence/proofs/a000224/OpHack/QuadraticResidue224/Target224.lean Endpoint lines: 33, 43, 50
S05 ferrers-hierarchy	Claim_Evidence/proofs/ferrers/OpHack/Ferrers/DefectExtension.lean Endpoint lines: 104
S06 ferrers-hierarchy	Claim_Evidence/proofs/ferrers/OpHack/Ferrers/YoungDiagramDefect.lean Endpoint lines: 8, 42
S07 ferrers-hierarchy	Claim_Evidence/proofs/ferrers/OpHack/Ferrers/YoungDiagramRestrictedPrefix.lean Endpoint lines: 158
S08 erdos169-fourap	Claim_Evidence/proofs/erdos169-fourap/OpHack/MainResult.lean Endpoint lines: 17
S09 opdp13-quartic	Claim_Evidence/proofs/opdp13/OpHack/PowerSemigroup/Evaluation.lean Endpoint lines: 116
S10 opdp13-quartic	Claim_Evidence/proofs/opdp13/OpHack/PowerMoment/Bound.lean Endpoint lines: 86

Source and claim	Included source and endpoint lines
S11 opdp13-quartic	Claim_Evidence/proofs/opdp13/OpHack/PowerFourier/Growth.lean Endpoint lines: 59
S12 opdp13-quartic	Claim_Evidence/proofs/opdp13/OpHack/PowerUpper/Function.lean Endpoint lines: 9
S13 opdp13-quartic	Claim_Evidence/proofs/opdp13/OpHack/PowerUpper/LeastIndex.lean Endpoint lines: 40
S14 opdp43-halfturn	Claim_Evidence/proofs/opdp43/OpHack/NoThree/Main.lean Endpoint lines: 8, 56
S15 erdos1060-uniform-partial	Claim_Evidence/proofs/erdos1060-uniform-partial/src/SigmaFiberAsymptotic.lean Endpoint lines: 38, 49
S16 erdos1060-uniform-partial	Claim_Evidence/proofs/erdos1060-uniform-partial/src/EntropyConversion.lean Endpoint lines: 45
S17 erdos21-sharp-residual	Claim_Evidence/proofs/erdos21-sharp-residual/OpHack/Erdos21/Main.lean Endpoint lines: 25, 33, 41, 49, 56
S18 erdos944-order-bound	Claim_Evidence/proofs/erdos944-order-bound/OpHack/Erdos944Bounds.lean Endpoint lines: 28, 42, 54
S19 opdp89-parity-local-maxima	Claim_Evidence/proofs/opdp89-parity-sharp/SharpClassification.lean Endpoint lines: 22, 33, 40
S20 erdos829-log-five-thirds	Claim_Evidence/proofs/erdos829-log-five-thirds/FiveThirdsCanonical.lean Endpoint lines: 8, 14